

POTLI: On demand Luggage storage network

Project Proposal for UCS503P

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August 18, 2026

1 Higher-order goal

This project aims to make short-distance urban travel a little less stressful by giving people a safe, nearby place to leave their bags whenever they don't want to lug them around – whether they're killing time before a hotel check-in, fitting in some sightseeing between train connections, or just running errands across an unfamiliar city. While the broader theme is about smoothing out everyday urban travel and opening up a genuine extra income stream for small local businesses, the immediate engineering objective is to translate this idea into a measurable, deployable software solution.

2 Time-to-value

- We will deliver meaningful increments quickly by prototyping the core booking-and-handoff workflow and validating it with a small set of real partner shops within a short iteration window.
- We will prioritize fast feedback over perfection by using weekly iterations (and targeting CI-backed automation early) to reduce release friction.
- Success is defined by what can be delivered and measured in the first iteration, then improved based on partner and traveler feedback.

3 Problem Statement

Travelers and short-term visitors in Indian cities – tourists between check-in and check-out, students home for the day, professionals in town for a single meeting – regularly need somewhere safe to leave a bag for a few hours, but have almost nowhere reliable to turn. Common pain points include:

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- **Scarcity:** official lockers exist only at a handful of railway stations and airports, and are frequently full, expensive, or out of service.
- **Low discoverability:** even when a nearby shop or business might be willing to hold a bag, travelers have no easy way to find or trust that option.
- **Trust and safety:** leaving belongings with a stranger, with no verification, receipt, or accountability, is understandably uncomfortable.
- **Untapped local income:** small shop and business owners often have a spare corner, shelf, or storeroom sitting idle, with no simple way to earn from it.

Why this matters now (contextual relevance): as more people travel for work, tourism, and short visits, and as tier-2/3 cities see rising footfall without matching storage infrastructure, even a lightweight coordination layer between travelers and local businesses can noticeably reduce this everyday friction.

4 Proposed Solution

4.1 Overview

Build a **web-based** “Storage-on-Demand” network with the following components:

- **Traveler discovery & booking portal:** search nearby partner shops, see capacity/price/hours, and book a slot in advance or on the spot.
- **Partner onboarding portal:** shop and business owners list their available space, set capacity, pricing, and working hours, and manage incoming bookings.
- **Booking & handoff workflow:** QR-code based check-in and check-out so both sides can verify exactly what was dropped off and what was picked up.
- **Trust & safety layer:** verified partner listings, photo confirmation at drop-off, and post-pickup ratings.
- **Payments & payouts:** the traveler pays through the platform; the partner receives their share automatically once the booking is completed.
- **Admin/ops dashboard:** city-wise coverage, partner performance, and revenue summaries.

4.2 Core workflow

1. Traveler searches for storage near a location and picks a partner and a time window.
2. System generates a booking ID and a QR code for that booking.
3. Partner scans/confirmes the bag at drop-off; the bag is tagged with the booking ID.
4. Traveler returns within the window and the partner scans/confirmes pickup.
5. Payment is settled and the partner’s payout is credited.

4.3 Operational constraints for an educational, tier-2/3 setting

- Keep the solution usable on low-to-mid range devices, since many partner shop owners will not have high-end smartphones.
- Keep partner onboarding extremely simple – most partners will be small shopkeepers, not tech-savvy businesses.
- Avoid dependence on advanced research algorithms; focus on workflow correctness, trust, and reliable handoffs.
- Design for deployability using common web hosting and managed databases.

5 Solution Approach

This is an engineering course project, so we focus on scalable coordination and trust-related workflow aspects rather than novel algorithm research.

5.1 Partner listing & duplicate-signup checks

Duplicate/quality checks on partner listings will be heuristic-based:

- Normalize partner details (business name, locality/address, phone number).
- Use a simple similarity measure (e.g., name + locality token overlap) to flag likely duplicate signups, without claiming any state-of-the-art matching technique.
- Show flagged listings to an admin for review; never auto-approve or auto-reject a partner without a human check.

5.2 Web architecture

A typical 3-tier web architecture:

- **Frontend:** responsive UI for two user types – travelers (search, book, track) and partners (list space, manage bookings).
- **Backend API:** authentication, location-based search, booking and handoff state transitions, payments.
- **Database:** users, partners, listings, bookings, handoff logs, and transactions.

5.3 CI/CD and fast delivery enabler

CI/CD will be used to:

- Automatically build and run tests on every change.
- Produce repeatable deployment artifacts to staging.
- Enable safe and frequent releases of incremental features (e.g., shipping QR-based handoff before layering on payments).

6 Evaluation Criterion

We define success using measurable metrics tied to the core booking-and-handoff workflow.

6.1 Primary evaluation metric

Booking Fulfillment Time (BFT): time between a traveler starting a search and receiving a confirmed booking at a partner location.

- Target: median BFT ≤ 3 minutes during the pilot window.
- Attribution: measured from database timestamps (search/request vs. confirmed booking).

6.2 Secondary evaluation metrics

- **Handoff success rate:** percentage of bookings where both drop-off and pickup are confirmed without disputes.
- **Partner utilization:** percentage of a partner’s listed capacity actually booked per week.
- **Partner payout turnaround:** time between booking completion and payout being credited.
- **Traveler satisfaction:** post-pickup 1–5 rating on ease and trust.
- **Reliability:** platform availability for search/booking during business hours (e.g., $\geq 99\%$ uptime in pilot).

6.3 Pilot validation plan

- Recruit 8–12 partner shops/businesses in a chosen locality (e.g., near a railway station or a busy market) and 15–20 traveler participants.
- Compare metrics against a baseline of no formal storage option (carrying bags around or informal arrangements), collected via short interviews before the pilot.

7 Scalability

The proposition should scale in both theoretical and practical senses.

1. **Scalable (theoretically):** The system should support growth in listings, bookings, and concurrent searches by:
 - decoupling components (frontend/backend),
 - using geo-indexed queries for “storage near me” search and pagination for results,
 - using stateless API design.
2. **Operationally deployable (practically):** The system should run on common web hosting:
 - managed database,
 - containerized or platform-hosted deployment,
 - CI/CD pipeline for staging/production.

8 Engine availability heuristic

A mature set of “engines” (tooling/services) is available to implement this as a web-based project:

- Web framework for REST APIs and responsive pages.
- Managed database with geospatial/location query support.
- Authentication approach (token-based or session-based) for both traveler and partner accounts.
- QR-code generation/scanning library for handoff verification.
- Payment gateway integration (e.g., UPI-based) for bookings and partner payouts.
- Notification channel (in-app, optionally SMS/WhatsApp) for booking confirmations and reminders.
- Test framework for unit/integration tests.
- CI runner and staging deployment target.

We will use these mature components to focus on workflow design, trust, and measurement rather than inventing new infrastructure.

9 Project scope and deliverables

9.1 Initial deliverable

- Traveler search + booking flow for nearby partners
- Partner onboarding + listing management (capacity, price, hours)
- QR-based drop-off and pickup handoff workflow
- Booking status tracking (Booked → Dropped → Stored → Picked up)
- Basic logging and timestamps for BFT measurement
- CI: build + backend unit tests + linting
- CD to staging with a single command or pipeline job

9.2 Subsequent deliverables

- Payments integration and automated partner payouts
- Ratings and reviews for partners
- Duplicate/quality checks for new partner listings
- Admin dashboard for coverage, utilization, and revenue metrics
- SMS/WhatsApp notifications for booking confirmations and reminders
- Improved geo-search, filtering, and indexed queries for scale readiness

10 Risks and mitigations

- **Trust and safety of stored items:** photo confirmation at drop-off/pickup, clear liability terms, verified partner onboarding, and an optional insurance add-on in a later iteration.
- **Partner adoption:** keep onboarding and day-to-day use extremely simple, minimize the steps needed to confirm a drop-off/pickup, and offer clear, transparent payout terms.
- **Payment/logistics risk:** hold payment until pickup is confirmed (escrow-style), rather than releasing it immediately at drop-off.
- **Evaluation measurement ambiguity:** instrument timestamps and define clear booking/handoff states before development begins.
- **Operational issues in pilot:** use a staging environment early and ensure CI/CD produces repeatable deployments.

11 Summary

This project proposes POTLI, a deployable web platform that connects travelers who need short-term luggage storage with local shop and business owners who have spare space to offer. It meets the course criteria by emphasizing time-to-value through rapid iterations, implementing CI/CD to support frequent safe releases, and using measurable evaluation criteria (especially booking fulfillment time). It stays focused on engineering workflow, trust, and scalability readiness rather than research-driven novelty, while creating a genuine extra income opportunity for local partners.